

colored blue. Therefore, we assume that the remaining board contains 20 blue squares, 21 black squares and 22 white squares.

If we could tile this board using straight triominoes, then we would use $63/3 = 21$ straight triominoes. These triominoes would cover 21 blue squares, 21 black squares and 21 white squares. This contradicts the fact that this board contains 20 blue squares, 21 black squares and 22 white squares. Therefore, we cannot tile this board using straight triominoes. $\square$

Example 23: The $3x+1$ Conjecture

Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x+1$. A famous conjecture, sometimes known as the $3x+1$ conjecture, states that for all positive integers x , when we repeatedly apply the transformation, we will eventually reach the integer 1.

For example, starting with $x=13$, we find $T(13) = 3 \cdot 13 + 1 = 40$, $T(40) = 40/2 = 20$, $T(20) = 20/2 = 10$, $T(10) = 10/2 = 5$, $T(5) = 3 \cdot 5 + 1 = 16$, $T(16) = 16/2 = 8$, $T(8) = 8/2 = 4$, $T(4) = 4/2 = 2$, $T(2) = 2/2 = 1$. The $3x+1$ conjecture has been verified using computers for all integers x up to 5.48×10^{18} .

See the article "J.C. Lagarias, The Ultimate Challenge: The $3x+1$ Problem, the American Mathematical Society, Providence, 2010." for a fascinating discussion of this problem and the results that have been found by mathematicians attacking it.

Some recommended books by George Polya

1. Mathematical Discovery, Vols 1. and 2
2. How to Solve It.
3. Mathematics and Plausible Reasoning